

Course 5283B

5 Credits: 4

Program Elective

Source-grounded

IoT Security

□ Understand the basics of IoT, sensors, actuators and IoT boards in real time

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTTR syllabus PDF for Course 5283B is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5283B.pdf from the uploaded official SITTTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Cyber Forensics And Information security
Course code	5283B
Course title	IoT Security
Semester	5 Credits: 4
Course category	Program Elective
Periods per week	4 (L:4T:0 P:0) Periods per semester:60
Periods per semester	60

Item	Official information
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

23 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- Understand the basics of IoT, sensors, actuators and IoT boards in real time environment. • Learn arduino programming
- Familiarize analog and digital communication
- Understand the Security requirements in IoT

Course outcomes

CO	Official outcome	Study level
CO1	15 Understanding and IoT boards in real time environment Understand the syntax of control statements, 14 Understanding	See official syllabus
CO2	operators , I/O functions for problem solving Understand analog and digital communications	See official syllabus
CO3	15 Understanding with arduino	See official syllabus
CO4	Understand the Security requirements in IoT 14 Understanding	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 2 2
CO2	CO2 2 2
CO3	CO3 - 1
CO4	CO4 1 3 3 1

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	real time environment	5	As specified
Module 2	for problem solving	6	As specified
Module 3	Understand analog and digital communications with arduino	6	As specified
Module 4	Understand the Security requirements in IoT	6	As specified

MODULE 1

real time environment

This module is presented from the official syllabus scope for IoT Security. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: real time environment
- Official duration: As specified
- Official topic entries captured: 5

- The full source excerpt is preserved below for coverage checking.

3 Understanding versus M2M, CPS, WoT; Enabled technologies, Networking

3 Understanding versus M2M, CPS, WoT; Enabled technologies, Networking is an official topic in Module 1 of IoT Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding versus M2M, CPS, WoT; Enabled technologies, Networking using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding versus m2m, cps, wot; enabled technologies, networking should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding versus M2M, CPS, WoT; Enabled technologies, Networking means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 1-3 Understanding versus M2M, CPS, WoT; Enabled technologies, Networking തിരിച്ചറിയൽ definition/meaning തിരിച്ചറിയൽ. തിരിച്ചറിയൽ തിരിച്ചറിയൽ, steps, application തിരിച്ചറിയൽ തിരിച്ചറിയൽ. Technical terms English- തിരിച്ചറിയൽ തിരിച്ചറിയൽ.

3 Understanding components, Challenges and applications Sensors: Definition, Characteristics,

3 Understanding components, Challenges and applications Sensors: Definition, Characteristics, is an official topic in Module 1 of IoT Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding components, Challenges and applications Sensors: Definition, Characteristics, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding components, challenges and applications sensors: definition, characteristics, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding components, Challenges and applications Sensors: Definition, Characteristics, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-3 Understanding components, Challenges and applications
Sensors: Definition, Characteristics, ഓപ്റ്റിക്കൽ സെൻസർ definition/meaning ഓപ്റ്റിക്കൽ സെൻസറുടെ പ്രവർത്തനം, steps, application ഓപ്റ്റിക്കൽ സെൻസറിന്റെ ഉപയോഗം. Technical terms English-ഐസ്ക്രീം ഓപ്റ്റിക്കൽ സെൻസർ.

Deviations, Types-Scalar, Multimedia, 3 Understanding Hybrid and virtual; Considerations

Deviations, Types-Scalar, Multimedia, 3 Understanding Hybrid and virtual; Considerations is an official topic in Module 1 of IoT Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Deviations, Types-Scalar, Multimedia, 3 Understanding Hybrid and virtual; Considerations using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, deviations, types-scalar, multimedia, 3 understanding hybrid and virtual; considerations should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Deviations, Types-Scalar, Multimedia, 3 Understanding Hybrid and virtual; Considerations means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- ∞ Deviations, Types-Scalar, Multimedia, 3 Understanding Hybrid and virtual; Considerations ∞ ∞ definition/meaning ∞ ∞ . ∞ ∞ ∞ , steps, application ∞ ∞ ∞ . Technical terms English- ∞ ∞ ∞ .

Actuators: Definition and Types 3 Understanding Classification of IoT boards:

Actuators: Definition and Types 3 Understanding Classification of IoT boards: is an official topic in Module 1 of IoT Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Actuators: Definition and Types 3 Understanding Classification of IoT boards: using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, actuators: definition and types 3 understanding classification of iot boards: should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Actuators: Definition and Types 3 Understanding Classification of IoT boards: means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 1-അക്ചുവേറ്റർ: Definition and Types 3

Understanding Classification of IoT boards: അക്ചുവേറ്റർ definition/meaning
അക്ചുവേറ്റർ. അക്ചുവേറ്റർ തരങ്ങൾ, steps, application അക്ചുവേറ്റർ
അക്ചുവേറ്റർ. Technical terms English-അക്ചുവേറ്റർ അക്ചുവേറ്റർ.

Microcontroller boards, Single board Understanding 3 controller, System on Chipboard CPS, IoT versus WoT; Enabled technologies, Networking components, Challenges and applications Sensors: Definition, Characteristics, Deviations, Types-Scalar, Multimedia, Hybrid and virtual; Considerations Actuators: Definition, Types-Hydraulic, Pneumatic, Electric, Thermal or Magnetic, Mechanical, Soft and shape memory polymers; Characteristics Classification of IoT boards: Microcontroller boards, Single board controller, System on Chipboard Understand the syntax of control statements, operators , I/O functions

Microcontroller boards, Single board Understanding 3 controller, System on Chipboard CPS, IoT versus WoT; Enabled technologies, Networking components, Challenges and applications Sensors: Definition, Characteristics, Deviations, Types-Scalar, Multimedia, Hybrid and virtual; Considerations Actuators: Definition, Types-Hydraulic, Pneumatic, Electric, Thermal or Magnetic, Mechanical, Soft and shape memory polymers; Characteristics Classification of IoT boards:

Microcontroller boards, Single board controller, System on Chipboard Understand the syntax of control statements, operators , I/O functions is an official topic in Module 1 of IoT Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Microcontroller boards, Single board Understanding 3 controller, System on Chipboard CPS, IoT versus WoT; Enabled technologies, Networking components, Challenges and applications Sensors: Definition, Characteristics, Deviations, Types-Scalar, Multimedia, Hybrid and virtual; Considerations Actuators: Definition, Types-Hydraulic, Pneumatic, Electric, Thermal or Magnetic, Mechanical, Soft and shape memory polymers; Characteristics Classification of IoT boards: Microcontroller boards, Single board controller, System on Chipboard Understand the syntax of control statements, operators , I/O functions using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.

- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, microcontroller boards, single board understanding 3 controller, system on chipboard cps, iot versus wot; enabled technologies, networking components, challenges and applications sensors: definition, characteristics, deviations, types-scalar, multimedia, hybrid and virtual; considerations actuators: definition, types-hydraulic, pneumatic, electric, thermal or magnetic, mechanical, soft and shape memory polymers; characteristics classification of iot boards: microcontroller boards, single board controller, system on chipboard understand the syntax of control statements, operators , i/o functions should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Microcontroller boards, Single board Understanding 3 controller, System on Chipboard CPS, IoT versus WoT; Enabled technologies, Networking components, Challenges and applications Sensors: Definition, Characteristics, Deviations, Types-Scalar, Multimedia, Hybrid and virtual; Considerations Actuators: Definition, Types-Hydraulic, Pneumatic, Electric, Thermal or Magnetic, Mechanical, Soft and shape memory polymers; Characteristics Classification of IoT boards: Microcontroller boards, Single board controller, System on Chipboard Understand the syntax of control statements, operators , I/O functions means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- Microcontroller boards, Single board Understanding 3 controller, System on Chipboard CPS, IoT versus WoT; Enabled technologies, Networking components, Challenges and applications Sensors: Definition, Characteristics, Deviations, Types-Scalar, Multimedia, Hybrid and virtual; Considerations Actuators: Definition, Types-Hydraulic, Pneumatic, Electric, Thermal or Magnetic, Mechanical, Soft and shape memory polymers; Characteristics Classification of IoT boards: Microcontroller boards, Single board controller, System on Chipboard Understand the syntax of control statements, operators , I/O functions definition/meaning .

രീതികൾ, steps, application രീതികൾ രീതികൾ. Technical terms English-
രീതികൾ രീതികൾ.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

real time environment

	Introduction: Introduction, Evolution, IoT	
M1.01		3
Understanding	versus M2M, CPS, WoT; Enabled technologies, Networking	
M1.02		3
Understanding	components, Challenges and applications Sensors: Definition, Characteristics, Deviations, Types-Scalar, Multimedia,	3
M1.03		
Understanding	Hybrid and virtual; Considerations Actuators: Definition and Types	3
M1.04		
Understanding	Classification of IoT boards: Microcontroller boards, Single board	
M1.05		3
Understanding	controller, System on Chipboard	
	Contents: Introduction: Introduction to IoT, Evolution of IoT-IoT versus M2M, IoT versus CPS, IoT versus WoT; Enabled technologies, Networking components, Challenges and applications Sensors: Definition, Characteristics, Deviations, Types-Scalar, Multimedia, Hybrid and virtual; Considerations	

Module study routine: Read the source wording, make a one-page concept map,
practise one short answer and one structured answer, then review the Malayalam
support notes.

MODULE 2

for problem solving

This module is presented from the official syllabus scope for IoT Security. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: for problem solving
- Official duration: As specified
- Official topic entries captured: 6
- The full source excerpt is preserved below for coverage checking.

Introduction to arduino

Introduction to arduino is an official topic in Module 2 of IoT Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Introduction to arduino using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, introduction to arduino should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Introduction to arduino means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-ഇൻ്റ്രൊഡക്ഷൻ ടു അർഡൂയ്നോ ഫൈർസെറ്റ് ഫൈർസെറ്റ്
definition/meaning ഫൈർസെറ്റ് ഫൈർസെറ്റ്. ഫൈർസെറ്റ് ഫൈർസെറ്റ് ഫൈർസെറ്റ്, steps, application
ഫൈർസെറ്റ് ഫൈർസെറ്റ് ഫൈർസെറ്റ്. Technical terms English-ഇൻ്റ്രൊഡക്ഷൻ ടു അർഡൂയ്നോ ഫൈർസെറ്റ്.

Program elements

Program elements is an official topic in Module 2 of IoT Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Program elements using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, program elements should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Program elements means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-എന്ന Program elements നോക്കി നിങ്ങളുടെ definition/meaning നോക്കി. നിങ്ങളുടെ നിങ്ങളുടെ, steps, application നോക്കി നിങ്ങളുടെ നിങ്ങളുടെ. Technical terms English-എന്ന നിങ്ങളുടെ നിങ്ങളുടെ.

statements

statements is an official topic in Module 2 of IoT Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify statements using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, statements should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What statements means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-എന്ന statements നോക്കി നിങ്ങളുടെ നിങ്ങളുടെ definition/meaning നോക്കി നിങ്ങളുടെ നിങ്ങളുടെ, steps, application നോക്കി നിങ്ങളുടെ നിങ്ങളുടെ. Technical terms English-എന്ന നിങ്ങളുടെ നിങ്ങളുടെ.

Connecting an external led

Connecting an external led is an official topic in Module 2 of IoT Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Connecting an external led using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, connecting an external led should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Connecting an external led means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-ഇംഗ്ലീഷ് കണക്റ്റിംഗ് ചെയ്യുന്ന ഒരു ബാഹ്യ ലെഡ് ലൈറ്റ് ബൾബിന്റെ
ഇംഗ്ലീഷ് definition/meaning നൽകുക. ഇംഗ്ലീഷ് ഇതിന്റെ ഉപയോഗം, steps,
application നൽകുക. Technical terms English-ൽ നൽകുക.
ഇംഗ്ലീഷ്.

Working with buttons

Working with buttons is an official topic in Module 2 of IoT Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Working with buttons using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, working with buttons should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Working with buttons means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- Working with buttons
definition/meaning . , steps, application
 . Technical terms English- .

Interrupts 2 Understanding Contents :Programming with Arduino: Introduction to arduino, Features, Components, Arduino IDE, Program elements-Structure, Variables and constants, Data types, Operators, Control statements, Loops, Functions, Arrays, String objects; Time, I/O function, Display, Random numbers, GPIO, Controlling LEDs-Blinking led without delay, Connecting an external led, RGB LED, The 7-segment display; Working with buttons-Connecting a button, Button with no resistor, The toggle switch, Button to serial, Button multiplexing; Interrupts

Interrupts 2 Understanding Contents :Programming with Arduino: Introduction to arduino, Features, Components, Arduino IDE, Program elements-Structure, Variables and constants, Data types, Operators, Control statements, Loops, Functions, Arrays, String objects; Time, I/O function, Display, Random numbers, GPIO, Controlling LEDs-Blinking led without delay, Connecting an external led, RGB LED, The 7-segment display; Working with buttons-Connecting a button, Button with no resistor, The toggle switch, Button to serial, Button multiplexing; Interrupts is an official topic in Module 2 of IoT Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Interrupts 2 Understanding Contents :Programming with Arduino: Introduction to arduino, Features, Components, Arduino IDE, Program elements-Structure, Variables and constants, Data types, Operators, Control statements, Loops, Functions, Arrays, String objects; Time, I/O function, Display, Random numbers, GPIO, Controlling LEDs-Blinking led without delay, Connecting an external led, RGB LED, The 7-segment display; Working with buttons-Connecting a button, Button with no resistor, The toggle switch, Button to serial, Button multiplexing; Interrupts using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, interrupts 2 understanding contents :programming with arduino: introduction to arduino, features, components, arduino ide, program elements-structure, variables and constants, data types, operators, control statements, loops, functions, arrays, string objects; time, i/o function, display, random numbers, gpio, controlling leds-blinking led without delay, connecting an external led, rgb led, the 7-segment display; working with buttons-connecting a button, button with no resistor, the toggle switch, button to serial, button multiplexing; interrupts should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Interrupts 2 Understanding Contents :Programming with Arduino: Introduction to arduino, Features, Components, Arduino IDE, Program elements-Structure, Variables and constants, Data types, Operators, Control statements, Loops, Functions, Arrays, String objects; Time, I/O function, Display, Random numbers, GPIO, Controlling LEDs-Blinking led without delay, Connecting an external led, RGB LED, The 7-segment display; Working with buttons-Connecting a button, Button with no resistor, The toggle switch, Button to serial, Button multiplexing; Interrupts means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-ഇററു Interrupts 2 Understanding Contents :Programming with Arduino: Introduction to arduino, Features, Components, Arduino IDE, Program elements-Structure, Variables and constants, Data types, Operators, Control statements, Loops, Functions, Arrays, String objects; Time, I/O function, Display, Random numbers, GPIO, Controlling LEDs-Blinking led without delay, Connecting an external led, RGB LED, The 7-segment display; Working with buttons-Connecting a button, Button with no resistor, The toggle switch, Button to serial, Button multiplexing; Interrupts ഇററു definition/meaning ഇററു. ഇററു ഇററു, steps, application ഇററു ഇററു. Technical terms English-ഇററു ഇററു.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

for problem solving		
M2.01	Introduction to arduino	2
Understanding		
M2.02	Program elements	2
Understanding		
M2.03	statements	2
Understanding		
M2.04	Connecting an external led	3
Understanding		
M2.05	Working with buttons	3
Understanding		
M2.06	Interrupts	2
Understanding		
	Series Test – I	1

Contents :Programming with Arduino: Introduction to arduino, Features, Components, Arduino IDE, Program elements-Structure, Variables and constants, Data types, Operators, Control statements, Loops, Functions, Arrays, String objects; Time, I/O function, Display, Random numbers, GPIO, Controlling LEDs-Blinking led without delay, Connecting an external led, RGB LED, The 7-segment display; Working with buttons-Connecting a button, Button with no resistor, The toggle switch, Button to serial, Button multiplexing; Interrupts

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Understand analog and digital communications with arduino

This module is presented from the official syllabus scope for IoT Security. Use the topic cards for learning and the source transcript for exact coverage.

As specified

CO-linked

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Understand analog and digital communications with arduino
- Official duration: As specified
- Official topic entries captured: 6
- The full source excerpt is preserved below for coverage checking.

Introduction-Serial communication

Introduction-Serial communication is an official topic in Module 3 of IoT Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Introduction-Serial communication using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, introduction-serial communication should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Introduction-Serial communication means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-ഇൻ്റ്രൊഡക്ഷൻ-സീരിയൽ കമ്മ്യൂണിക്കേഷൻ

സീരിയൽ കമ്മ്യൂണിക്കേഷൻ ന്റെ definition/meaning എന്താണ്. സീരിയൽ കമ്മ്യൂണിക്കേഷൻ ന്റെ steps, application എന്താണ്. Technical terms English-ൽ എഴുതുക.

Introduction Parallel communication

Introduction Parallel communication is an official topic in Module 3 of IoT Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Introduction Parallel communication using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, introduction parallel communication should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Introduction Parallel communication means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-പ്രാരംഭം Introduction Parallel communication

പരസ്പരം സമാന്തരമായി സംഭാഷണം നടത്തുന്നതിന്റെ definition/meaning എന്താണ്. പരസ്പരം സമാന്തരമായി സംഭാഷണം, steps, application എങ്ങനെ നടത്തണം. Technical terms English-ൽ എങ്ങനെ വിശദീകരിക്കണം.

Interfacing LCD character display

Interfacing LCD character display is an official topic in Module 3 of IoT Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Interfacing LCD character display using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, interfacing lcd character display should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Interfacing LCD character display means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-ഓ Interfacing LCD character display

ഓഓഓഓഓഓഓഓ ഓഓഓ definition/meaning ഓഓഓഓഓഓഓഓ. ഓഓഓഓ ഓഓഓഓ ഓഓഓഓഓ,
steps, application ഓഓഓഓ ഓഓഓഓഓഓഓ ഓഓഓഓ. Technical terms English-ഓ ഓഓഓഓ
ഓഓഓഓഓഓഓഓ.

UART Communication

UART Communication is an official topic in Module 3 of IoT Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify UART Communication using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, uart communication should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What UART Communication means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-ഓർട്ടർ UART Communication ഓർട്ടർ ഓർട്ടർ definition/meaning ഓർട്ടർ. ഓർട്ടർ ഓർട്ടർ ഓർട്ടർ, steps, application ഓർട്ടർ ഓർട്ടർ ഓർട്ടർ. Technical terms English-ഓർട്ടർ ഓർട്ടർ.

I2C Communication

I2C Communication is an official topic in Module 3 of IoT Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify I2C Communication using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, i2c communication should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What I2C Communication means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-ഇ I2C Communication ഇതിന്റെ നിർവ്വചനം/അർത്ഥം, steps, application ഇതിന്റെ പ്രയോഗം. Technical terms English-ഇ ഇതിന്റെ നിർവ്വചനം.

SPI Communication 3 Understanding Contents : Analog and Digital Communication with Arduino: Introduction-Serial communication, Parallel communication, Interfacing LCD character display UART Communication: UART protocol, Serial communication, Interfacing ESP8266 module I2C Communication: I2C protocol, ADXL345 module, Interfacing BMP180 module SPI Communication: SPI protocol, SD card interfacing, Ethernet module interfacing

SPI Communication 3 Understanding Contents : Analog and Digital Communication with Arduino: Introduction-Serial communication, Parallel communication, Interfacing LCD character display UART Communication: UART protocol, Serial communication, Interfacing ESP8266 module I2C Communication: I2C protocol, ADXL345 module, Interfacing BMP180 module SPI Communication: SPI protocol, SD card interfacing, Ethernet module interfacing is an official topic in Module 3 of IoT Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify SPI Communication 3 Understanding Contents : Analog and Digital Communication with Arduino: Introduction-Serial communication, Parallel communication, Interfacing LCD character display UART Communication: UART protocol, Serial communication, Interfacing ESP8266 module I2C Communication: I2C protocol, ADXL345 module, Interfacing BMP180 module SPI Communication: SPI protocol, SD card interfacing, Ethernet module interfacing using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, spi communication 3 understanding contents : analog and digital communication with arduino: introduction-serial communication, parallel communication, interfacing lcd character display uart communication: uart protocol,

serial communication, interfacing esp8266 module i2c communication: i2c protocol, adxl345 module, interfacing bmp180 module spi communication: spi protocol, sd card interfacing, ethernet module interfacing should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What SPI Communication 3 Understanding Contents : Analog and Digital Communication with Arduino: Introduction-Serial communication, Parallel communication, Interfacing LCD character display UART Communication: UART protocol, Serial communication, Interfacing ESP8266 module I2C Communication: I2C protocol, ADXL345 module, Interfacing BMP180 module SPI Communication: SPI protocol, SD card interfacing, Ethernet module interfacing means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-എസ് പി കമ്മ്യൂണിക്കേഷൻ 3 Understanding Contents : Analog and Digital Communication with Arduino: Introduction-Serial communication, Parallel communication, Interfacing LCD character display UART Communication: UART protocol, Serial communication, Interfacing ESP8266 module I2C Communication: I2C protocol, ADXL345 module, Interfacing BMP180 module SPI Communication: SPI protocol, SD card interfacing, Ethernet module interfacing
 ഏകദേശം 1000 വാക്കുകളിൽ നിർവ്വചന/അർത്ഥം എഴുതുക. ഏകദേശം 1000 വാക്കുകളിൽ, steps, application എഴുതുക. Technical terms English-ൽ എഴുതുക.
 ഏകദേശം 1000 വാക്കുകളിൽ.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

Understand analog and digital communications with arduino

M3.01	Introduction-Serial communication	2
Understanding		

M3.02	Introduction Parallel communication	2
Understanding		

M3.03	Interfacing LCD character display	2
Understanding		

M3.04	UART Communication	3
Understanding		

M3.05	I2C Communication	3
Understanding		

M3.06	SPI Communication	3
Understanding		

Contents : Analog and Digital Communication with Arduino: Introduction-Serial communication, Parallel communication, Interfacing LCD character display

UART Communication: UART protocol, Serial communication, Interfacing ESP8266 module

I2C Communication: I2C protocol, ADXL345 module, Interfacing BMP180 module

SPI Communication: SPI protocol, SD card interfacing, Ethernet module interfacing

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Understand the Security requirements in IoT

This module is presented from the official syllabus scope for IoT Security. Use the topic cards for learning and the source transcript for exact coverage.

As specified

CO-linked

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Understand the Security requirements in IoT
- Official duration: As specified
- Official topic entries captured: 6
- The full source excerpt is preserved below for coverage checking.

2 Understanding Architecture Security Architecture in the Internet of

2 Understanding Architecture Security Architecture in the Internet of is an official topic in Module 4 of IoT Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding Architecture Security Architecture in the Internet of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding architecture security architecture in the internet of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding Architecture Security Architecture in the Internet of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 4-2 Understanding Architecture Security

Architecture in the Internet of പരിപാടി പരിപാടി definition/meaning

പരിപാടി. പരിപാടി പരിപാടി, steps, application പരിപാടി പരിപാടി

പരിപാടി. Technical terms English-പരിപാടി പരിപാടി.

3 Understanding Things

3 Understanding Things is an official topic in Module 4 of IoT Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding Things using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding things should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding Things means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-3 Understanding Things ഏകദേശം 3000 വർഷം
definition/meaning എന്തെങ്കിലും വസ്തു. എന്തെങ്കിലും വസ്തു, steps, application
എന്തെങ്കിലും വസ്തു. Technical terms English-3000 വർഷം.

Security Requirements in IoT 2 Understanding Insufficient Authentication /

Security Requirements in IoT 2 Understanding Insufficient Authentication / is an official topic in Module 4 of IoT Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Security Requirements in IoT 2 Understanding Insufficient Authentication / using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, security requirements in iot 2 understanding insufficient authentication / should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Security Requirements in IoT 2 Understanding Insufficient Authentication / means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

2 Understanding Authorization Vulnerabilities – Secrecy and Secret-

2 Understanding Authorization Vulnerabilities – Secrecy and Secret- is an official topic in Module 4 of IoT Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding Authorization Vulnerabilities – Secrecy and Secret- using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding authorization vulnerabilities – secrecy and secret- should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding Authorization Vulnerabilities – Secrecy and Secret- means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 4-2 Understanding Authorization Vulnerabilities – Secrecy and Secret- അപര്യം definition/meaning അപര്യം. അപര്യം അപര്യം അപര്യം, steps, application അപര്യം അപര്യം അപര്യം. Technical terms English- അപര്യം അപര്യം അപര്യം.

2 Understanding Key Capacity

2 Understanding Key Capacity is an official topic in Module 4 of IoT Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding Key Capacity using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding key capacity should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding Key Capacity means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-2 Understanding Key Capacity $\square\square\square\square\square\square\square\square\square$
 $\square\square\square\square$ definition/meaning $\square\square\square\square\square\square\square\square\square\square$. $\square\square\square\square$ $\square\square\square\square$ $\square\square\square\square\square\square$, steps,
application $\square\square\square\square$ $\square\square\square\square\square\square\square\square$ $\square\square\square\square$. Technical terms English- $\square$ $\square\square\square\square$
 $\square\square\square\square\square\square\square\square$.

Transport Encryption 3 Understanding Security Requirements in IoT Architecture - Security in Enabling Technologies - Security Concerns in IoT Applications. Security Architecture in the Internet of Things - Security Requirements in IoT - Insufficient Authentication/Authorization - Insecure Access Control - Threats to Access Control, Privacy, and Availability - Attacks Specific to IoT. Vulnerabilities - Secrecy and Secret-Key Capacity - Authentication/Authorization for Smart Devices - Transport Encryption - Attack & Fault trees Text / Reference T/R Book Title/Author Sudeep Mishra, Anandarupmukherjee and Arijit Roy, Introduction to IoT, New T1 Delhi: University Cambridge Press, 2021. (Chapter 4) Cornel Amariei, Arduino Development Cook Book, Birmingham: Packt T2 Publishing Ltd., 2015. (Chapter 2-6) Fei HU, "Security and Privacy in Internet of Things (IoTs): Models, Algorithms, R1 and Implementations", CRC Press, 2016 Russell, Brian and Drew Van Duren, "Practical Internet of Things Security", Packt R2 Publishing, 2016. Online Resources Sl.No Website Link 1 <https://www.arduino.cc/en/IoT/HomePage> 2 <https://learn.adafruit.com/category/internet-of-things-iot>

Transport Encryption 3 Understanding Security Requirements in IoT Architecture - Security in Enabling Technologies - Security Concerns in IoT Applications. Security Architecture in the Internet of Things - Security Requirements in IoT - Insufficient Authentication/Authorization - Insecure Access Control - Threats to Access Control, Privacy, and Availability - Attacks Specific to IoT. Vulnerabilities - Secrecy and Secret-Key Capacity - Authentication/Authorization for Smart Devices - Transport Encryption - Attack & Fault trees Text / Reference T/R Book Title/Author Sudeep Mishra, Anandarupmukherjee and Arijit Roy, Introduction to IoT, New T1 Delhi: University Cambridge Press, 2021. (Chapter 4) Cornel Amariei, Arduino Development Cook Book, Birmingham: Packt T2 Publishing Ltd., 2015. (Chapter 2-6) Fei HU, "Security and Privacy in Internet of Things (IoTs): Models, Algorithms, R1 and Implementations", CRC Press, 2016 Russell, Brian and Drew Van Duren, "Practical Internet of Things Security", Packt R2 Publishing, 2016. Online Resources Sl.No Website Link 1 <https://www.arduino.cc/en/IoT/HomePage> 2 <https://learn.adafruit.com/category/internet-of-things-iot> is an official topic in Module 4 of IoT Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Transport Encryption 3 Understanding Security Requirements in IoT Architecture - Security in Enabling Technologies - Security Concerns in IoT Applications. Security Architecture in the Internet of Things - Security Requirements in IoT - Insufficient Authentication/Authorization - Insecure Access Control - Threats to Access Control, Privacy, and Availability - Attacks Specific to IoT. Vulnerabilities – Secrecy and Secret-Key Capacity - Authentication/Authorization for Smart Devices - Transport Encryption – Attack & Fault trees Text / Reference T/R Book Title/Author Sudeep Mishra, Anandarupmukherjee and Arijit Roy, Introduction to IoT, New T1 Delhi: University Cambridge Press, 2021. (Chapter 4) Cornel Amariei, Arduino Development Cook Book, Birmingham: Packt T2 Publishing Ltd., 2015. (Chapter 2-6) Fei HU, “Security and Privacy in Internet of Things (IoTs): Models, Algorithms, R1 and Implementations”, CRC Press, 2016 Russell, Brian and Drew Van Duren, “Practical Internet of Things Security”, Packt R2 Publishing, 2016. Online Resources Sl.No Website Link 1 <https://www.arduino.cc/en/IoT/HomePage> 2 <https://learn.adafruit.com/category/internet-of-things-iot> using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, transport encryption 3 understanding security requirements in iot architecture - security in enabling technologies - security concerns in iot applications. security architecture in the internet of things - security requirements in iot - insufficient authentication/authorization - insecure access control - threats to access control, privacy, and availability - attacks specific to iot. vulnerabilities – secrecy and secret-key capacity - authentication/authorization for smart devices - transport encryption – attack & fault trees text / reference t/r book title/author sudeep mishra, anandarupmukherjee and arijit roy, introduction to iot, new t1 delhi: university cambridge press, 2021. (chapter 4) cornel amariei, arduino development cook book, birmingham: packt t2 publishing ltd., 2015. (chapter 2-6) fei hu, “security and privacy in internet of things (iots): models, algorithms, r1 and implementations”, crc press, 2016 russell, brian and drew van duren, “practical internet of things security”, packt r2 publishing, 2016. online resources sl.no website link 1 <https://www.arduino.cc/en/iot/homepage> 2 <https://learn.adafruit.com/category/internet-of-things-iot> should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Aspect	What to prepare
Definition/identity	What Transport Encryption 3 Understanding Security Requirements in IoT Architecture - Security in Enabling Technologies - Security Concerns in IoT Applications. Security Architecture in the Internet of Things - Security Requirements in IoT - Insufficient Authentication/Authorization - Insecure Access Control - Threats to Access Control, Privacy, and Availability - Attacks Specific to IoT. Vulnerabilities – Secrecy and Secret-Key Capacity - Authentication/Authorization for Smart Devices - Transport Encryption – Attack & Fault trees Text / Reference T/R Book Title/Author Sudeep Mishra, Anandarupmukherjee and Arijit Roy, Introduction to IoT, New T1 Delhi: University Cambridge Press, 2021. (Chapter 4) Cornel Amariei, Arduino Development Cook Book, Birmingham: Packt T2 Publishing Ltd., 2015. (Chapter 2-6) Fei HU, “Security and Privacy in Internet of Things (IoTs): Models, Algorithms, R1 andImplementations”, CRC Press,2016 Russell, Brian and Drew Van Duren, “Practical Internet of Things Security”, Packt R2 Publishing, 2016. Online Resources SI.No Website Link 1 https://www.arduino.cc/en/IoT/HomePage 2 https://learn.adafruit.com/category/internet-of-things-iot means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-Transport Encryption 3 Understanding Security Requirements in IoT Architecture - Security in Enabling Technologies - Security Concerns in IoT Applications. Security Architecture in the Internet of Things - Security Requirements in IoT - Insufficient Authentication/Authorization - Insecure Access Control - Threats to Access Control, Privacy, and Availability - Attacks Specific to IoT. Vulnerabilities – Secrecy and Secret-Key Capacity - Authentication/Authorization for Smart Devices - Transport Encryption – Attack & Fault trees Text / Reference T/R Book Title/Author Sudeep Mishra, Anandarupmukherjee and Arijit Roy, Introduction to IoT, New T1 Delhi: University Cambridge Press, 2021. (Chapter 4) Cornel Amariei, Arduino Development Cook Book, Birmingham: Packt T2 Publishing Ltd., 2015. (Chapter 2-6) Fei HU, “Security and Privacy in Internet of Things (IoTs): Models, Algorithms, R1 andImplementations”, CRC Press,2016 Russell, Brian and Drew Van Duren, “Practical Internet of Things Security”, Packt R2 Publishing, 2016. Online Resources SI.No Website Link 1 <https://www.arduino.cc/en/IoT/HomePage> 2 <https://learn.adafruit.com/category/internet-of-things-iot> definition/meaning . steps, application . Technical terms English-

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

Understand the Security requirements in IoT		
Security Requirements in IoT		
M4.01		2
Understanding	Architecture	
	Security Architecture in the Internet of	
M4.02		3
Understanding	Things	
M4.03	Security Requirements in IoT	2
Understanding	Insufficient Authentication /	
M4.04		2
Understanding	Authorization	
	Vulnerabilities – Secrecy and Secret-	
M4.05		2
Understanding	Key Capacity	
M4.06	Transport Encryption	3
Understanding	Series Test – II	1
Security Requirements in IoT Architecture - Security in Enabling Technologies - Security		
Concerns in IoT Applications. Security Architecture in the Internet of Things - Security		
Requirements in IoT - Insufficient Authentication/Authorization - Insecure Access		

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

Definition → Principle → Components/steps → Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure → Observation → Result → Safety

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain 3 Understanding versus M2M, CPS, WoT; Enabled technologies, Networking. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding versus M2M, CPS, WoT; Enabled technologies, Networking*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 3 Understanding components, Challenges and applications Sensors: Definition, Characteristics,. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding components, Challenges and applications Sensors: Definition, Characteristics,.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Deviations, Types-Scalar, Multimedia, 3 Understanding Hybrid and virtual; Considerations. (3 marks)

Answer: Begin with the standard definition or identity of *Deviations, Types-Scalar, Multimedia, 3 Understanding Hybrid and virtual; Considerations.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Actuators: Definition and Types 3 Understanding Classification of IoT boards:.. (3 marks)

Answer: Begin with the standard definition or identity of *Actuators: Definition and Types 3 Understanding Classification of IoT boards:.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Module 2

Define or explain Introduction to arduino. (3 marks)

Answer: Begin with the standard definition or identity of *Introduction to arduino*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Program elements. (3 marks)

Answer: Begin with the standard definition or identity of *Program elements*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain statements. (3 marks)

Answer: Begin with the standard definition or identity of *statements*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Connecting an external led. (3 marks)

Answer: Begin with the standard definition or identity of *Connecting an external led*. State the governing principle, list the key components or stages, and close with one

relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Module 3

Define or explain Introduction-Serial communication. (3 marks)

Answer: Begin with the standard definition or identity of *Introduction-Serial communication*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Introduction Parallel communication. (3 marks)

Answer: Begin with the standard definition or identity of *Introduction Parallel communication*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Interfacing LCD character display. (3 marks)

Answer: Begin with the standard definition or identity of *Interfacing LCD character display*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps,

application application application application.

Define or explain UART Communication. (3 marks)

Answer: Begin with the standard definition or identity of *UART Communication*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: definition, principle/steps, application application application.

Module 4

Define or explain 2 Understanding Architecture Security Architecture in the Internet of. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding Architecture Security Architecture in the Internet of*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: definition, principle/steps, application application application.

Define or explain 3 Understanding Things. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding Things*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: definition, principle/steps, application application application.

Define or explain Security Requirements in IoT 2 Understanding Insufficient Authentication /. (3 marks)

Answer: Begin with the standard definition or identity of *Security Requirements in IoT 2 Understanding Insufficient Authentication /*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 2 Understanding Authorization Vulnerabilities - Secrecy and Secret-. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding Authorization Vulnerabilities - Secrecy and Secret-*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **3 Understanding versus M2M, CPS, WoT; Enabled technologies, Networking.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **Introduction to arduino**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Introduction-Serial communication**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **2 Understanding Architecture Security Architecture in the Internet of**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link <https://www.arduino.cc/en/IoT/HomePage>

- <https://learn.adafruit.com/category/internet-of-things-iot>

IN-PAGE SEARCH

Search this lesson

Prepared from the supplied official SITTTR syllabus PDF for Course 5283B. Verify institutional updates against the current official curriculum.